

$$G = (V, E)$$

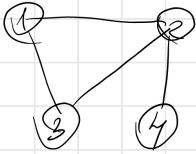
$$V = \{1, 2, \dots, n\}$$

$$E \subseteq V \times V =$$

$$= \{(u, v), u \in V, v \in V\}$$

Способы задания
графов:

1) рисунок



2) перечисление

$$V = \{1, 2, 3, 4, 5, 6\}$$

$$E = \{a = (1, 3), b = (1, 4), c = (1, 5)\}$$

3) матрица смежности

	1	2	3	4
1	1	0	1	0
2	1	0	1	0
3	0	1	0	0
4	0	1	0	1

4) матрица инцидентности

	a	b	c	d	e
1					
2					
3					
4					
5					

5) список смежности

$$1: 3, 4, 5$$

$$2: 3, 6$$

$$3: 2, 1$$

$$4: 1$$

$$5: 1$$

$$6: 2$$

✓ 4.1.

$1) 2^{(C_n^1 + n)}$

(x, y)

$V = \{1, 2, \dots, n\}$

2) $2 \cdot C_n^2 + n$

3) $2^{n^2 - n}$

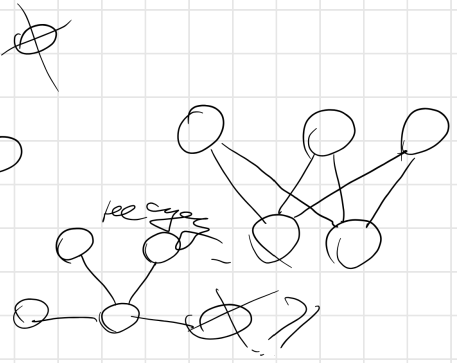
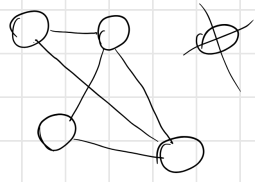
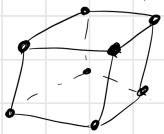
4) $2^{C_n^2}$

✓ 4.2.

1) C_n^2

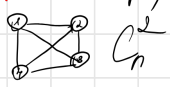
2) $2^{C_n^2}$

✓ 4.5



Типы графов:

1) K_n - полный граф



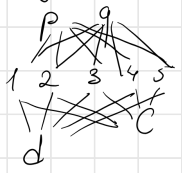
2) K_{pq} - полный двудольный граф.

$V = A \cup B \quad (A \cap B = \emptyset)$

(различимые объекты) $\{b_1, b_2, \dots, b_q\}$

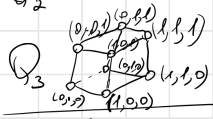
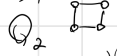
$\{a_1, a_2, \dots, a_p\}$

каждый с каждым



$p \cdot q$

3) Q_k - k-мерный куб



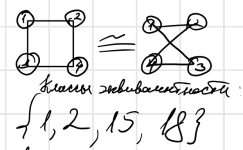
$w = (1, 1, 0, 1, 1)$
 $x = (1, 1, 1, 1, 1)$

$h(w, x) = 2$

Q_4^2

$$G_1 \cong G_2$$

$\Leftrightarrow$
 $\varphi: V_1 \rightarrow V_2 \Rightarrow (a, b) \in E_1 \Rightarrow$
 $\Rightarrow (\varphi(a), \varphi(b)) \in E_2$
пример



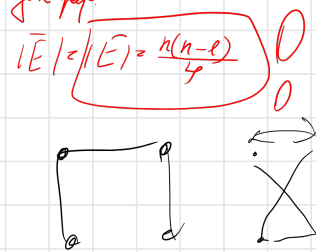
- $\{1, 2, 15, 18\}$
- $\{3, 14, 6, 8, 9, 11\}$
- $\{4, 13\}$
- $\{5, 7, 17\}$
- $\{10, 16, 12\}$

как-то абстрактно
 как графы:
 $m_{K_6} = C_6^2 = 15$
 $\sim 4, 15$

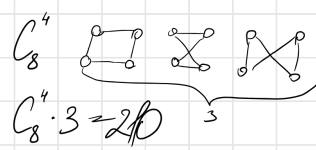
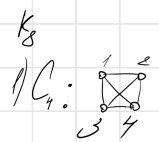
- $\{1, 3, 4\}$
- $\{2, 5, 6\}$

Граф G изоморфен
 самографическому,
 если $G \cong G$

пример
 если выпадает - есть
 граф



4.21



2) P_4

$C_8 \cdot \frac{4!}{2} = 84$



$C_8 \cdot 4$

4.22 $K_{3,7}$ - 10 вершин
 3 из них соединены
 со всеми



$C_3^2 \cdot C_7^2$

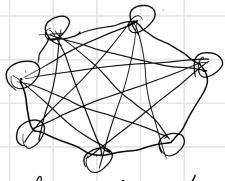
1) $C_4 | C_3^2 C_7^2 \cdot 4$

2) $K_{1,3} | C_3^1 C_7^3 + C_7^1$



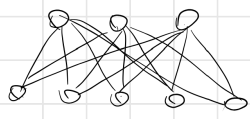
~4.23

1) K_7



$$C_7^1 \cdot C_6^1 \cdot C_5^1 \cdot C_4^1 \cdot C_3^1$$

2) $K_{3,5}$

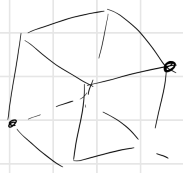


$$C_5^3$$

~4.24



2



K_4

$ecc(x)$ = расстояние до самой удалённой точки

радиус графа - min из $ecc(x)$

диаметр - max из $ecc(x)$

центр - м.во вершины, у которых $\{x / ecc(x) = \text{радиус}\}$

Теорема

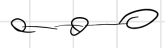
В связном $G \Rightarrow$ справедливо:
 радиус $\leq$ диаметр
 диаметр $\leq 2 \cdot$ радиус

~4.25

1) P_2

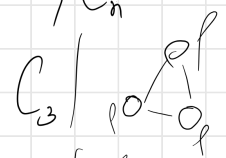


P_3



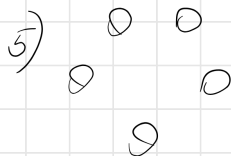
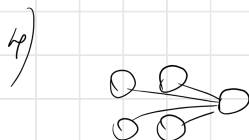
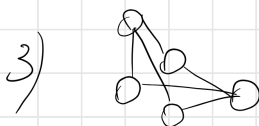
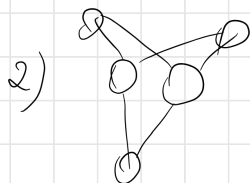
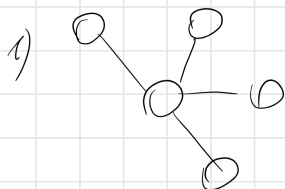
$$\begin{aligned} \text{diam}(P_n) &= n-1 \\ \text{rad}(P_n) &= \lfloor \frac{n-1}{2} \rfloor \end{aligned}$$

2) C_n



$$\begin{aligned} \text{rad} &= n-2 \\ \text{diam} &= n-2 \end{aligned}$$

~4.28



~4.32

$$|V(G)| = n$$

x edges

$n-x$ edges

$$\max \left(C_x^2 - C_{n-x}^2 \right) =$$

$$= \max \left(\frac{x(x-1)}{2} + \frac{(n-x)(n-x-1)}{2} \right) =$$

$$= \max \left(\frac{1}{2} (x^2 - x + n^2 - xn - n - xn + x^2 - 2) \right) =$$

$$= \max \left(\frac{1}{2} (2x^2 + n^2 - 2x - 2xn - 2) \right) =$$

$$f(x) = 2x^2 - 2nx - n + n^2$$

$$|V(G)| = n$$

$$1 \leq x \leq n-1$$

$$f'(x) = 4x - 2n$$

$$f'(x) = 0$$

$$4x - 2n = 0$$

$$x = \frac{n}{2}$$

$$(x=1), (x=n-1)$$

$$f(x) = f(n-1) = 2 - 2n - n + n^2 =$$

$$= n^2 - 3n + 2$$

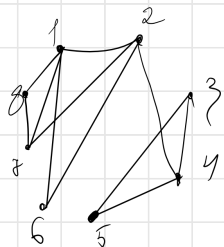
$$f\left(\frac{n}{2}\right) = \frac{1}{2} n^2 - n^2 - n + n^2 =$$

$$= \frac{1}{2} n^2 - n$$

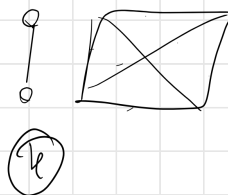
$$n^2 - 3n + 2 \geq \frac{1}{2} n^2 - n$$

$$\frac{n^2}{2} - 2n + 2 \geq 0$$

$\sqrt{4.33}$



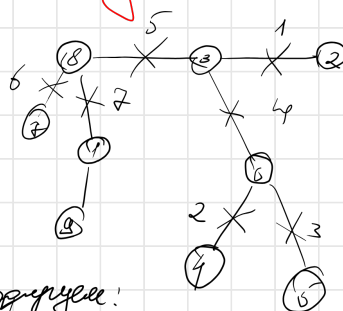
~ 4.35



4.36

- 1) 
- 2) 
- 3) 
- 4) 
- 5) 
- 6) 

Ког. Прютера



Задача:
м. Веревка
в колл м-2

$$P(+) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 2 & 6 & 6 & 3 & 8 & 8 & 1 \end{pmatrix}$$

Сравни:

[illegible]

$$P(T) = \underbrace{(3, 6, 6, 3, 8, 8, 1)}_{9 \text{ шт}}$$

$$\sim 4.51$$

$$13592$$



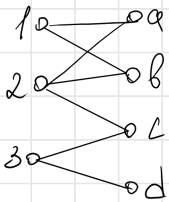
$$|V| - |E| + |F| = 2$$

$$|E| \leq 3|V| - 6$$

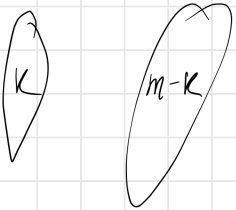
$$3|F| \leq |E|$$

$$G = (V, E) \text{ — графовый (диграф)}$$

$$V = V_1 \cup V_2$$



$$m \rightarrow \max$$



$$m = K(n - K)$$

$$n = 2 \cdot K \Rightarrow K = \frac{n}{2}$$

$$n = 2K + 1$$

$$B = \{0, 1\}$$

$$f(x_1, x_2, \dots, x_n)$$

$$f: B^n \rightarrow B$$

1)

x_1	x_2	$\dots$	x_n	f

2) $f(\underbrace{\dots}_{2^n})$

3) определен

$$f(x_1, x_2, \dots, x_n) = f(\tilde{x}_n) = \tilde{f}$$

$$|P_2(n)| = 2^{2^n}$$

~5.1(2)

$$f(x_1, x_2, x_3) = 1 \Leftrightarrow x_1 + x_2 + x_3 \equiv 1 \pmod{2}$$

x_1	x_2	x_3	f
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

$$f = (01101001)$$

$$2^3 = 8$$

$$1+1+1=3 \pmod{2}=1$$

$$f(x_1, x_2, x_3) = 1 \Leftrightarrow x_1 \leq x_2 \leq x_3$$

$$x_1 \leq x_3$$

x_1	x_2	x_3	f
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

$$f = (11110001)$$

~5.2.

1) $f(0, 0, 0) = 1$

x_1	x_2	x_3	f
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

$$\frac{2^3 - 1}{2}$$

x_1	x_2	x_3	f
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

5.3 $f(0, 0, \dots, 0) = f(1, 1, \dots, 1)$



$$2^{2^{n-1}}$$

$$2) 2^{2^n - C_n^k}$$

$$3) C_{2^n}^k$$



6) $\pi^2(p_1, p_2, \dots, p_n)$

$$f(x_1, x_2, \dots, x_n) = f(p_1, p_2, \dots, p_n)$$

$$2^{n+1}$$

$$f = (10001) \quad \tilde{g} = (00010)$$

$$h(x_1, x_2) = f(x_1, g(x_1, x_2))$$

x_1	x_2	f	g	h_1	h_2	h_3	h_4
0	0	1	0	1	0	0	
0	1	0	0	1	1	0	
1	0	0	1	1	0	0	
1	1	1	0	0	0	1	

$$h_2 = g(x_2, f(x_2, x_1))$$

$$h_3 = g(x_2, g(x_2, x_1))$$

25.4

3)

x_1	x_2	x_3	F	g	h
0	0	0	1	0	0
0	0	1	1	0	0
0	1	0	0	0	1
0	1	1	0	0	0
1	0	0	0	1	0
1	0	1	0	1	0
1	1	0	1	0	1
1	1	1	1	0	1

$$F(f(x_1, x_2), g(x_2, x_3))$$

1,

$$\tilde{f} = (10111011)$$

x_1	x_2	x_3	F
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

$$\begin{aligned}
 x \rightarrow y &= \bar{x} \vee y \\
 x \downarrow y &= \overline{x \vee y} \\
 x \oplus y &= x\bar{y} \vee \bar{x}y \\
 x \sim y &= xy \vee \bar{x}\bar{y} \\
 \overline{x \vee y} &= \bar{x} \wedge \bar{y} \\
 \overline{x \wedge y} &= \bar{x} \vee \bar{y} \\
 \bar{\bar{x}} &= x \oplus 1
 \end{aligned}$$

$$\begin{aligned}
 f(x_1, x_2) &= (x_2 \rightarrow x_1)(x_2 \downarrow x_2) = \\
 &= (\bar{x}_2 \vee x_1)(\bar{x}_2) = \\
 &= \bar{x}_2
 \end{aligned}$$

$$\begin{aligned}
 &((x_1 \vee x_2 \bar{x}_3) \sim (\bar{x}_1 \rightarrow \bar{x}_2 x_3)) (x_2 \downarrow x_3) = \\
 &= ((x_1 \vee x_2 \bar{x}_3) \sim (x_1 \vee \bar{x}_2 x_3)) (\overline{x_2 \vee x_3}) = \\
 &= ((x_1 \vee x_2 \bar{x}_3) \vee \overline{(x_1 \vee x_2 x_3)}) (\overline{x_2 \vee x_3}) = \\
 &= (x_1 \vee x_2 \bar{x}_3 \vee \bar{x}_1 (\bar{x}_2 \vee x_3)) (\bar{x}_2 \bar{x}_3) = \\
 &= (x_1 \vee x_2 \bar{x}_3 \vee \bar{x}_1 \bar{x}_2 \vee \bar{x}_1 x_3) (\bar{x}_2 \bar{x}_3) = \\
 &= x_1 \bar{x}_2 \bar{x}_3 \vee \bar{x}_1 \bar{x}_2 \bar{x}_3 = \bar{x}_2 \bar{x}_3
 \end{aligned}$$

$$\begin{aligned}
 3) f(x_1, x_2, x_3) &= ((x_1 \oplus x_2) \rightarrow x_3) (\overline{x_3 \rightarrow x_2}) = \\
 &= ((\overline{x_1 \oplus x_2}) \vee x_3) (\overline{x_3 \vee x_2}) = \\
 &= ((\bar{x}_1 \bar{x}_2 \vee \bar{x}_1 x_2) \vee x_3) (\bar{x}_3 \bar{x}_2) = \\
 &= ((\bar{x}_1 \bar{x}_2) (\bar{x}_1 x_2) \vee x_3) (\bar{x}_3 \bar{x}_2) = \\
 &= ((\bar{x}_1 \vee x_2) (\bar{x}_1 \vee \bar{x}_2) \vee x_3) (\bar{x}_3 \bar{x}_2) = \\
 &= (\bar{x}_1 \bar{x}_2 \vee x_2 \bar{x}_1 \vee x_3) (\bar{x}_3 \bar{x}_2) = \\
 &= \bar{x}_1 \bar{x}_2 \bar{x}_3 \vee x_3 \bar{x}_2 = \bar{x}_2 \bar{x}_3 (\bar{x}_1 \vee 1)
 \end{aligned}$$

$f(x_1, x_2)$
 одне реп аргумент.

vb.8

$$f(x_1, x_2, x_3)$$

a) $2^{3-1} = 16$ x_3 - ~~переменна~~

b) $2^{3-2} = 4$ x_1, x_2 - ~~переменні~~

в) $2^3 - 3 \cdot 2^2 + C_3 \cdot 2^2 - 2^0$
 $\underbrace{\quad}_{x_1 \text{ перемає } 1}$ $\underbrace{\quad}_{x_1 \wedge x_2 \text{ перемає } 1}$ $\underbrace{\quad}_{x_1 \wedge x_2 \wedge x_3 \text{ перемає } 1}$
 $\underbrace{\quad}_{x_2 \text{ перемає } 1}$ $\underbrace{\quad}_{x_1 \wedge x_2 \text{ перемає } 1}$ $\underbrace{\quad}_{x_1 \wedge x_2 \text{ перемає } 1}$

vb.

$x_1 x_2 x_3$	$x_1 \vee x_2 \vee x_3$	$x_1 \oplus x_2 \oplus x_3$		
0 0 0	0	0	0	0
0 0 1	1	1	1	1
0 1 0	1	1	1	1
0 1 1	1	0	1	1
1 0 0	1	1	0	0
1 0 1	1	0	1	0
1 1 0	1	0	0	0
1 1 1	1	1	0	1

АИФ, САНФ, КИФ
СКНФ

$$K_1 \vee K_2 \vee \dots \vee K_n$$

$$xy \vee x\bar{z}$$

$$f(x, y, z) = xy \vee x\bar{z} = \text{АИФ}$$

$$= xy(z \vee \bar{z}) \vee x(y \vee \bar{y})\bar{z} =$$

$$= xy z \vee xy \bar{z} \vee x y \bar{z} \vee x \bar{y} \bar{z} = \text{САНФ}$$

б. 13

$$1) x_1(x_2 | x_3) = \bar{x}_1 \vee (\bar{x}_2 x_3) = \bar{x}_1 \vee \bar{x}_2 \vee x_3 = \text{АИФ, СКНФ}$$

$$2) (x_1 \sim x_2) \rightarrow (x_2 \sim x_3) =$$

$$= (x_1 x_2 \vee \bar{x}_1 \bar{x}_2) \rightarrow (x_2 x_3 \vee \bar{x}_2 \bar{x}_3) =$$

$$= (\bar{x}_1 \bar{x}_2 \wedge \bar{x}_1 \bar{x}_2) \vee (x_2 x_3 \vee \bar{x}_2 \bar{x}_3) =$$

$$= ((\bar{x}_1 \vee \bar{x}_2) \wedge (x_1 \vee x_2)) \vee (x_2 x_3 \vee \bar{x}_2 \bar{x}_3)$$

$$= \bar{x}_1 x_2 \vee x_1 \bar{x}_2 \vee x_2 x_3 \vee \bar{x}_2 \bar{x}_3 = \text{АИФ}$$

$$= x_2 (\bar{x}_1 \vee x_3) \vee \bar{x}_2 (x_1 \vee \bar{x}_3) =$$

$$AB \vee CD = (A \vee CD)(B \vee CD) =$$

$$= (A \vee C)(A \vee D)(B \vee C)(B \vee D)$$

б. 13

$$6) \overline{(x_1 \sim x_2)} \oplus (x_1 \downarrow x_3) =$$

$$= x_1 \vee x_3 \vee \bar{x}_1 \bar{x}_2 x_3 =$$

$$= x_1 \bar{x}_2 \vee \bar{x}_1 x_2 x_3 \vee \bar{x}_1 \bar{x}_2 \bar{x}_3 = \text{АИФ}$$

$$= (x_1 \vee x_3)(\bar{x}_1 \vee \bar{x}_2)(x_2 \vee \bar{x}_3)(\bar{x}_2 \vee x_3) = \text{КИФ}$$

$$f = (01110010)$$

x_1	x_2	x_3	f
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

Спрощен САНФ

$$\bar{x}_1 \bar{x}_2 x_3 \vee \bar{x}_1 x_2 \bar{x}_3 \vee \bar{x}_1 x_2 x_3 \vee x_1 x_2 \bar{x}_3$$

Спрощен СКНФ

x_1	x_2	x_3	f
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

$$f = (x_1 \vee x_2 \vee x_3)(\bar{x}_1 \vee x_2 \vee x_3) \cdot (\bar{x}_1 \bar{x}_2 \bar{x}_3) / (x_1 x_2 x_3)$$

нб.11

$$\begin{aligned}
 1) f &= x \oplus y \quad g = x \sim y = xy \vee \bar{x}\bar{y} \\
 f^* &= \overline{x \oplus y} = \overline{xy \vee \bar{x}\bar{y}} = \\
 &= \overline{xy} \wedge \overline{\bar{x}\bar{y}} = (\overline{xy}) / (\bar{x}\bar{y}) = \\
 &= xy \vee \bar{x}\bar{y} = x \sim y
 \end{aligned}$$

$$2) f = \overline{x|y} = \overline{\bar{x}\bar{y}} = \overline{\bar{x} \vee \bar{y}} = xy$$

$$\begin{aligned}
 3) f &= (\bar{x} \rightarrow \bar{y}) \rightarrow (y \rightarrow x) \\
 g &= (x \rightarrow y) \cdot (\bar{y} \rightarrow \bar{x})
 \end{aligned}$$

$$\begin{aligned}
 f^* &= \overline{(\bar{x} \rightarrow \bar{y}) \rightarrow (y \rightarrow \bar{x})} = \\
 &= \overline{(\bar{x} \vee y) \rightarrow (y \vee \bar{x})} = \\
 &= \overline{(x\bar{y}) \vee (y \vee \bar{x})} = \\
 &= \overline{(\bar{x} \vee y)(\bar{y} \vee x)} = \\
 &= 0
 \end{aligned}$$

нб.12

$$\begin{aligned}
 f &= (x \vee y \vee z)(y \oplus z) \vee xyz \\
 f^* &= (xyz) \vee (y \sim z) \wedge (x \vee y \vee z)
 \end{aligned}$$

$$\overline{\bar{x}\bar{y}} = \overline{(x\bar{y})} = (xy)$$

$$f = x \vee y \quad f^* = x | y$$

Полином Хемминга

$$\begin{aligned}
 f &= \alpha_0 \oplus \alpha_1 x_1 \oplus \dots \oplus \alpha_n x_n \oplus \\
 &\oplus \alpha_{12} x_1 x_2 \oplus \alpha_{13} x_1 x_3 \oplus \dots \oplus \alpha_{ij} x_i x_j \oplus \dots \\
 &\dots \oplus \alpha_{123\dots n} x_1 x_2 \dots x_n
 \end{aligned}$$

нб.20

$$\begin{aligned}
 1) (x_1 \sim x_2) \oplus x_3 &= \\
 = (x_1 x_2 \vee \bar{x}_1 \bar{x}_2) \oplus x_3 &= \\
 = \overline{(x_1 x_2 \vee \bar{x}_1 \bar{x}_2)} \oplus x_3 &= \\
 = (\bar{x}_1 \bar{x}_2 \vee x_1 x_2) \oplus x_3 &= \\
 = ((x_1 x_2 \oplus 1)(\bar{x}_1 \bar{x}_2 \oplus 1) \oplus 1) \oplus x_3 &= \\
 = (x_1 x_2 \oplus 1)(x_1 x_2 \oplus x_1 \bar{x}_2 \oplus \bar{x}_1 x_2 \oplus 1) \oplus x_3 &= \\
 = x_1 x_2 \oplus \bar{x}_1 \bar{x}_2 \oplus x_1 x_2 \oplus x_1 \bar{x}_2 \oplus \bar{x}_1 x_2 \oplus \bar{x}_1 \bar{x}_2 \oplus 1 \oplus x_3 &= \\
 = x_1 \oplus x_2 \oplus x_3 \oplus 1
 \end{aligned}$$

н.5.20

$$\begin{aligned} b) f &= x_1 x_2 \rightarrow (x_3 \sim x_4) = \\ &= f = \overline{(x_1 x_2 \vee (\bar{x}_3 \bar{x}_4 \vee x_3 x_4))} = \\ &= \overline{(x_1 \vee x_2)(\bar{x}_3 \vee x_4)(\bar{x}_3 \vee x_4)} = \\ &= \overline{(x_1 x_2 (\bar{x}_3 \bar{x}_4 \vee x_3 x_4))} = \\ &= x_1 x_2 (x_3 \sim x_4) \end{aligned}$$

$K_1 \vee K_2 \vee \dots \vee K_n$ - САНФ

$K_1 \oplus K_2 \oplus \dots \oplus K_n$

САНФ: $x \vee y = x \oplus y$

АНФ: $x \vee y = xy \oplus x \oplus y$

н.5.21

x_1	x_2	x_3	f
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0

АНФ: $\bar{x}_1 \bar{x}_2 \bar{x}_3 \vee \bar{x}_1 \bar{x}_2 x_3 \vee \bar{x}_1 x_2 \bar{x}_3$

$$\begin{aligned} \vee x_1 x_2 x_3 &= (x_1 \oplus 1)(x_2 \oplus 1)(x_3 \oplus 1) \dots (x_1)(x_2 \oplus 1)(x_3) = \\ &= 1 \oplus x_2 \oplus x_1 x_3 \oplus x_2 x_3 \end{aligned}$$

Теперь получаем каноническую форму коэф. и сравним

$$\begin{aligned} F &= \alpha_0 + \alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3 + \\ &+ \alpha_{12} x_1 x_2 + \alpha_{13} x_1 x_3 + \alpha_{23} x_2 x_3 + \\ &+ \alpha_{123} x_1 x_2 x_3 \end{aligned}$$

$$f(0,0,0) = \alpha_0 + 1 \Rightarrow \alpha_0 = 1$$

из ради. уст.

$$f(0,0,1) = \alpha_0 + \alpha_3 = 1 \Rightarrow \alpha_3 = 0$$

A

[A] - ил. до всех элементов

$$A = \{\bar{x}, xy, xy\bar{y}\} \quad \left. \begin{array}{l} \text{наим.} \\ \text{смет.} \end{array} \right\}$$

$$A = \{x \oplus y, xy\bar{y}, 1\}$$

Если A, B - системы
гс-ил

(при этом A - полная)
то B будет полной, когда
 $\forall f \in B$ выражается через
A.

№ 6.1

$$\{ \bar{x} \}$$

Ответ: $\bar{x}_1, \bar{x}_2, x_1, x_2$

$$2) \{ \bar{0}, \bar{x} \}$$

Ответ: $0, \bar{x}_1, \bar{x}_2, 1$

$$3) \{ x_1, x_2, x_1 \oplus x_2 \}$$

Ответ: $\{ x_1, x_2, x_1 \oplus x_2, \}$

$$x_1 \oplus x_2 \oplus x_2 = x_1, x_1,$$

$$x_1 \oplus x_2 \oplus x_2 = x_1, x_1, x_1 \vee x_1 \bar{x}_2 \bar{x}_2$$

№ 6.2

$$f = \bar{x}, A = \{ \bar{0}, x \rightarrow y \}$$

$$\bar{x} = x \rightarrow 0$$

$$2) f = x, A = \{ x \oplus y \}$$

$$x = x \oplus (x \oplus y) = x$$

$$3) f = x \vee y, A = \{ x/y \}$$

$$\{ x/x/y/y \} = \bar{x} \bar{y}$$

$$4) f = x \oplus y$$

$$A = \{ \bar{0}, x \rightarrow y \}$$

$$x \oplus y$$

$$\bar{x} \bar{y} \vee x \bar{y}$$

$$(x \rightarrow y) \rightarrow (y \rightarrow x) \rightarrow 0$$

$$y \bar{x}$$

№ 6.5

$$1) f \in T_0, T_1$$

$$2^{2^n - 2}$$

$$2) 2^{2^n - 2}$$

$$3) f \in T_0 \otimes T_0$$

$$2^{2^n - 1}$$

$$4) (T_0 \vee T_1)$$

$$2^{2^n - 2} + 2^{2^n}$$

$$\boxed{2^{2^n} - 2^{2^n - 2} = 2^{2^n} (1 - \frac{1}{4})}$$

$$= \frac{3}{4} 2^{2^n}$$

Characterization

$$f \in S \Leftrightarrow \exists \alpha \in \{x, y\} \\ \Rightarrow \{x, y\}$$

L, M - sets of points

$f \in L$:

$$f = \alpha_0 \oplus \alpha_1 x \oplus \dots \oplus \alpha_n x^n \\ \alpha_i \in \{0, 1\}$$

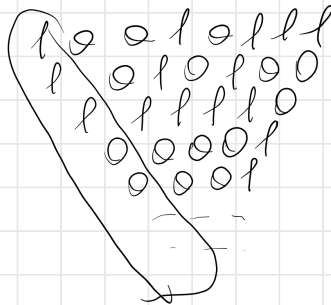
Ex. 1.2

$$f = (10010110)$$

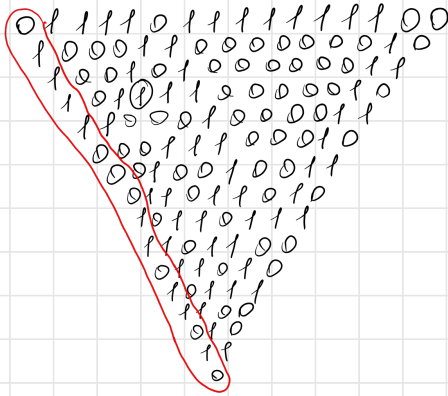
$x_1 x_2 x_3$	f				
000	1	1	1	1	1
001	0	0	0	1	x_3
010	0	0	1	1	x_2
011	1	1	1	0	
100	0	1	1	1	x_1
101	1	1	1	0	
110	1	1	0	0	
111	0	1	0	0	

$$\overline{x_1} \overline{x_2} \overline{x_3} \cup \overline{x_1} x_2 x_3 \cup x_1 \overline{x_2} x_3 \cup \\ \cup x_1 x_2 \overline{x_3}$$

$$1 \oplus x_3 \oplus x_2 \oplus x_1$$



6)
f₂



6.13

$$f_2 = x_1 \oplus x_1 x_2 \oplus x_2 x_2$$

$$f_1 = 0, f_2 = 1, f_3 = x_1 \oplus \bar{x}_1$$

$$f_4 = \bar{x}_2, f_5 = 1 \oplus x_1 \oplus x_2 = x_1 \sim x_2$$

$$f_6 = 1 \oplus x_1 \oplus x_2, f_7 = x_1, f_8 = x_2$$

$$2^3 = 8$$

6.14

$$1) (x_1 \sim x_2) \sim (x_3 \sim x_4)$$

$$(1 \oplus x_1 \oplus x_2) \sim (1 \oplus x_3 \oplus x_4) =$$

$$= 1 \oplus 1 \oplus x_1 \oplus x_2 \oplus 1 \oplus x_3 \oplus x_4 =$$

$$= 1 \oplus x_1 \oplus x_2 \oplus x_3 \oplus x_4 \in L$$

$$\begin{aligned} 2) (x_1 | \bar{x}_2) | (\bar{x}_1 | x_2) &= \\ &= (x_1 \bar{x}_2 \oplus 1) | (\bar{x}_1 x_2 \oplus 1) = \\ &= x_1 \bar{x}_2 \oplus x_1 \bar{x}_2 \oplus \bar{x}_1 x_2 \oplus \bar{x}_1 x_2 = \\ &= 1 \end{aligned}$$

6.14(5)

$$(x_1 \rightarrow x_2) \oplus (x_2 \rightarrow x_1) =$$

$$= (\bar{x}_1 \vee x_2) \oplus (\bar{x}_2 \vee x_1) =$$

$$= (x_1 \oplus 1) \vee x_2 \oplus (\bar{x}_2 \oplus 1) \vee x_1 =$$

$$= x_1 \oplus 1 \oplus x_2 \oplus 1 \vee x_2 \oplus x_2 \oplus 1 \vee \\ \vee x_1 \oplus 1 \oplus x_1 \vee x_2 \oplus x_1 =$$

$$= x_1 \oplus x_2$$

$$x \vee y = xy \oplus x \oplus y$$

$$|L_0|$$

$$f \in L \cap \overline{L_0} \Leftrightarrow \begin{cases} f \in L \\ f \notin L_0 \end{cases} \Leftrightarrow$$

$$f \in M \Leftrightarrow \forall \alpha, \beta$$

$$(\alpha \leq \beta) \Rightarrow f(\alpha) \leq f(\beta)$$

$$\Leftrightarrow \begin{cases} f = \alpha_0 \oplus \alpha_1 x_1 \oplus \dots \oplus \alpha_n x_n \\ f(0, \dots, 0) = 0 \Rightarrow \alpha_0 = 0 \end{cases}$$

$$\Downarrow$$

$$f = \alpha_1 x_1 \oplus \alpha_2 x_2 \oplus \dots$$

$$\oplus \alpha_n x_n, \alpha_i \in \{0, 1\}$$

$$|SL| \quad f \in SL \Leftrightarrow \begin{cases} f \in S \\ f \in L \end{cases}$$

$$f = f^* = \overline{\alpha_0 \oplus \alpha_1 x_1 \oplus \alpha_2 x_2 \oplus \dots \oplus \alpha_n x_n}$$

$$f = \alpha_0 \oplus \alpha_1 x_1 \oplus \alpha_2 x_2 \oplus \dots \oplus \alpha_n x_n$$

$$\alpha_0 \oplus \alpha_1 x_1 \oplus \alpha_2 x_2 \oplus \dots \oplus \alpha_n x_n =$$

$$= \alpha_0 \oplus \alpha_1 x_1 \oplus 1 \oplus \dots \oplus \alpha_n x_n \oplus 1$$

i.e. n+1 wires

v6.17

$$f = (00010011)$$

x_1	x_2	x_3	f
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

$$f = (00110001) \notin M$$

v6.18

$$0, 1, x_1, x_2, \vee, \wedge$$

6.25
 $A = \{x_1, x_2, x_3, x_4\}$
 f_1 f_2

	T_0	T_1	S	M	L
f_1	+	+	-	+	-
f_2	-	+	+	-	+

- 1) 1 +
- 2) $\bar{x}$ -
- 3) x_1, x_2 -
- 4) $x_1 x_2 x_3 x_4 \oplus x_1 x_2 x_3 \oplus 1$
- 5) $x_1 \oplus x_2 \oplus x_3$ +

6.26
 $FG \bar{T}_0 \cup \bar{T}_1 \cup S$
 f - ~~неизвестно~~
 $\{f, f, f\} = P_2$

T_0	T_1	S	M	L
-	-	-	-	-

Задача:
1) $f \notin T_0, f \notin T_1, f \notin S$
 $f(0, \dots, 0) = 1$
 $f(1, \dots, 1) = 0 \Rightarrow f \notin M$

2) α представлено
 n $f \in L$
 $f = \alpha_0 \alpha_1 \alpha_2 \dots \alpha_n$
 $f(0, 0, \dots, 0) = 1$
 $\alpha_0 = 0$

Проверим
 $f(x) = f(\bar{x})$
 $f = 1 \oplus x_1 \alpha_2 \oplus \dots \oplus \alpha_n = 0$
 $x_1 \oplus x_2 \oplus \dots \oplus \alpha_n = 1$
 $\bar{f} = 1 \oplus 1 \oplus \alpha_2 \oplus \dots \oplus \alpha_n$

$f(x) = 1 \oplus x_1 \oplus x_2 \oplus x_3$
 $f \in S$
 $?!$

	T_0	T_1	S	M	L
f_1	+	-	-	+	+
f_2	-	-	+	-	+

- 1) $\bar{x}$ +
- 2) $x_1 \sim x_2 = x_1 \oplus x_2 \oplus 1$ +
 $\bar{x}_1 x_2 \cup x_1 \bar{x}_2$
- 3) $x_1 \rightarrow x_2 = 1 \oplus x_1 \oplus x_2$ $\notin L$
- 4) +
- 5) +

6.28

1) $\{0, \bar{x}, x_1 \oplus x_2 \oplus x_3, (x_1 \vee x_2)(x_1 \vee x_3)(x_2 \vee x_3)\}$

База

1) $LA S = P_2$, A - ~~нормальная система~~

2) $\forall f \in A$:
 $\{A / \{f\}\} \neq P_2$
База замкнута
 $\text{не для } f \text{ и } \bar{f}$

$\nwarrow$ База
 $\{f_1, f_2, f_3\}$

6.37
 $|P_2(n) - (T_0 \cup T_2 \cup S)| =$
 $2^{2^n - 1} + 2^{2^n - 1} + 2^{2^n - 1}$
 $+$

Home work

№6.25

$$[\underbrace{\{x_1, x_2\}}_{f_1}, \underbrace{\{x_1 \oplus x_2 \oplus 1\}}_{f_2}]$$

	$\bar{f}_0$	$\bar{f}_1$	S	M	L
f_1	+	+	-	+	-
f_2	-	+	-	-	+

1) $\perp$ +

2) $\bar{x}$ -

3) x_1/x_2 -

4) $x_1 x_2 x_3 x_4 \oplus x_1 x_2 x_3 \oplus 1$ +

5) $x_1 \oplus x_2 \oplus x_3$ -

№6.29

1) $\{f_1=0, f_2=1, f_3=\bar{x},$
 $f_4=x(y \oplus z) \oplus yz\}$

базис $\{f_1, f_2, f_3\}$

2) базис $\{f_1, f_2\}$

7.3/2

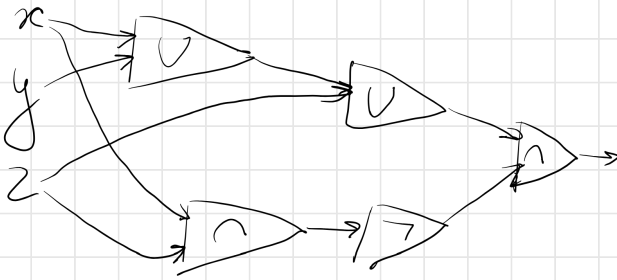
Загрубование

$$x\bar{z} \vee y\bar{z} \vee \bar{x}y \vee \bar{x}z$$

$$y(\bar{z} \vee \bar{x}) \vee \underbrace{x\bar{z} \vee \bar{x}z}_{x \oplus z} =$$

$$= y(\bar{z} \vee \bar{x}) \vee (x \vee z)(\bar{x} \vee \bar{z}) =$$

$$= (\bar{z} \vee \bar{x})(y \vee x \vee z)$$

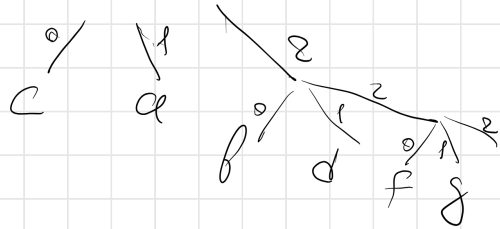


1811

$$3) P = 0.2; 0.1; 0.3; 0.1; 0.05; 0.1; 0.05; 0.1$$

$$A = \{g, b, z\}$$

$$P = \frac{1}{5}, \frac{1}{10}, \frac{3}{10}, \frac{1}{10}, \frac{1}{20}, \frac{1}{10}, \frac{1}{20}, \frac{1}{10}$$



$$C - 0$$

$$a - 1$$

$$b - 20$$

$$d - 21$$

$$f - 220$$

$$g - 221$$

$$C(P, V) = 6,9$$

Договор
Купор-н

 $\sim 8.13/4)$

$$P = (0,2 \quad 0,20 \quad 0,55)$$

aa
ab
ac
ba
bb
bc
ca
cb
cc

$$\begin{cases} a \rightarrow 0 \\ b \rightarrow 10 \\ c \rightarrow 11 \end{cases}$$

$$P(a) = (0,2 \cdot 0,2) = 0,04$$

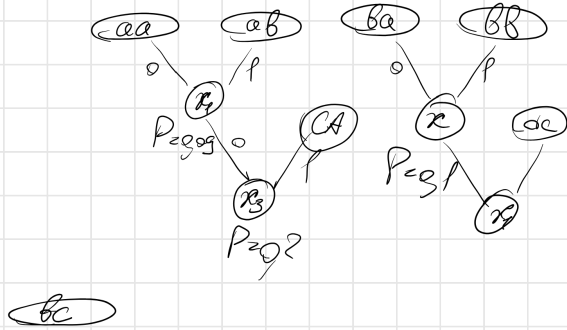
$$P(a|b) = P(b|a) = 0,2 \cdot 0,25 = 0,05$$

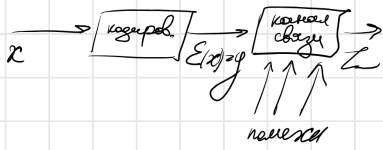
$$P(a) = P(a) = 0,2 \cdot 0,55 = 0,11$$

$$P(00)_2 = 0,55 \cdot 0,55 =$$

$$z = 0,3025$$

aa	0,04
ab	0,05
ac	0,11
ba	0,05
bb	0,076
bc	
ca	
cb	
cc	





(001000) → (001010)

↖
введен шб

Алгоритм

Хэмминга

$$x = x_1 x_2 \dots x_n \rightarrow y = x_1 x_2 x_1 \dots x_n x_n x_1$$

$$01 \rightarrow 000111$$

$$E(x) = y$$

$$00 \rightarrow 000000$$

10

111000

$$x = 10$$

код-е

$$E(x) = 111000$$

канал связи

111010



$$J = \sum_i |x_i - y_i|$$

$$V = (x_1, x_2, \dots, x_n)$$

$$u = (y_1, \dots, y_n)$$

Теорема

ког исправляет t ошибок

$$d(V) \geq 2t+1$$

Теорема 2

ког обнаруживает t ошибок

$$d(V) \geq t+1$$

Пример

$$V = \{111000, 10101, 01110\}$$

1) когда раскр-е ? $d(V) = \min_{i \neq j} J(V_i, V_j)$

2) каково число ког обна-
 руживает ? $t=2$

3) $\sim$ исправляет ? $J(V_1, V_2) = J(111000, 10101)$

Пример

~8.15

$$\{1111100110011, 0011111\}$$

$$1.) \int (V_1, V_2) \int (V_2, V_1) =$$

$$= 4$$

$$2) \int (V_1, V_3) = 4$$

$$3) \int (V_2, V_3) = 4$$

$$\Rightarrow d(V) = 4$$

$$4 \geq t + p$$

$$3 \geq t$$

$$4 \geq 2t + p$$

$$3 \geq 2t$$

$$t \geq \frac{3}{2}$$

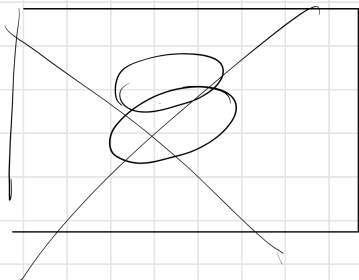
$$B_t(y) = \{z \in \{0, p\}^2\}$$

$$p(xy) \leq t y$$

$$B_p(y) = \{z \in \{0, p\}^2 \mid p(xy) \leq p\}$$

$$|B_1(y)| = k + p$$

$$y = (y_1, y_2, \dots, y_k)$$

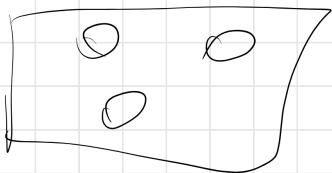


Has repetit.

Has galimius

betas

arab



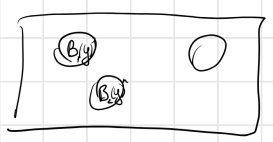
$$x = x_1 x_2 \dots x_k \rightarrow y_1 y_2 \dots y_k$$

$k < n$

$$n = 11 \Rightarrow k_{\max} =$$

$$x_1 x_2 \dots x_{11} \rightarrow y_1 y_2 \dots y_{k_{\max}}$$

$k = 7$



$n-1$ шагов
 2^k
 $\uparrow$
 количество элементов
 $B_r(y^{(i)})$

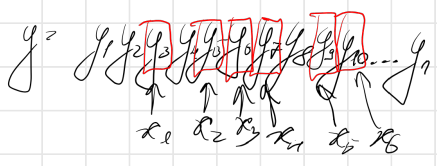
$2^k(n+1) \leq 2^n$

$\downarrow$

$n+1 \leq 2^{n-k}$

n -гара; k -регр.

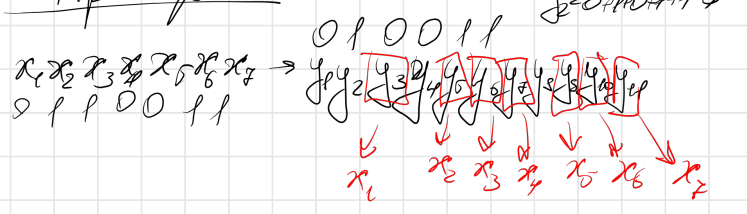
Найти $k_{\max}$ при котором
 возможен рекурсивный



$i \in 2^5 \rightarrow$ неформализованное
 разложение

$i \in 2^5$ проверочное
 разложение

Пример



$$y_1 y_2 y_3 y_4 y_5 y_6 y_7$$

$M_i = \{P \mid \text{взвешенное разложение}$
 которого i -ый раз $= 1\}$

$$1 = 1 \cdot 2^0 + 0 \cdot 2^1 + 0 \cdot 2^2 + \dots + 0 \cdot 2^n$$

	1	2	3	4	5	6	7	8	9	10	11
1	0	1	0	1	0	1	0	1	0	1	0
2	0	1	1	0	0	1	1	0	0	1	1
3	0	0	0	1	1	1	1	0	0	0	0
4	0	0	0	0	0	0	0	1	1	1	1

Пример

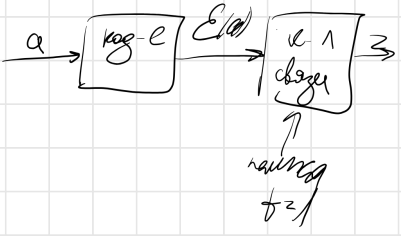
$$M_0 = \{1, 3, 5, 7, 9, 11, \dots\}$$

$$y = y_3 y_4 y_5 y_7$$

Пример

$$z = (110111101000)$$

$$a = ?$$



$$110111101000$$

$$z_1 z_2 z_3 z_4 z_5 z_6 z_7 z_8 z_9 z_{10} z_{11}$$

$$|a| = \int_{\delta y \in B} a^k$$

$$|z| = pp$$

$$\frac{n}{n}$$

$$l+n \leq 2^{n-k}$$

$$M_0 = \{1, 3, 5, 7, 9, 11\}$$

$$M_1 = \{2, 3, 6, 7, 10, 11\}$$

$$M_2 = \{4, 5, 6, 7\}$$

$$M_3 = \{8, 9, 10, 11\}$$

$$z_1 + z_3 + z_5 + z_7 + z_9 + z_{11} = 1 + 0 + 1 + 1 + 1 + 0 = 4$$

$$z_1 + z_3 + z_5 + z_7 + z_9 + z_{11}$$

$$= 1 + 0 + 1 + 1 + 0 + 0 = 3$$

$$z_1 + z_3 + z_5 + z_7 = 0$$

$$z_8 + z_9 + z_{10} + z_{11} = 1$$

$$(0, 1, 0, 1)$$

↓
10 не прав


~~~~~  
01101001011

$$|a| = |p| = k$$

$$|z| = |b| = n$$

$$n+k \leq 2^{n-k}$$